

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I LOCHANA SRI R S/ 192521348 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lochana Sri R S

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

ANSWERS

1. Cloud Architecture for Video Streaming:

Cloud Architecture:

The streaming platform should use a cloud-native architecture consisting of users, DNS, Content Delivery Network (CDN), Load Balancer, Auto-Scaling Application Servers, Object Storage, Distributed Database, and Monitoring Services. Video files are stored in scalable object storage while frequently accessed content is cached in CDN edge servers located closer to users. This reduces latency, improves streaming speed, and provides a better viewing experience even during peak traffic.

Load Balancing and Auto-Scaling:

A load balancer distributes incoming requests equally among multiple application servers, preventing any single server from becoming overloaded. Auto-scaling continuously monitors CPU utilization, memory usage, and network traffic. When user demand increases, additional virtual machines or containers are launched automatically. During low demand, unused resources are removed to reduce cloud costs while maintaining performance.

Caching Strategy:

Caching should be implemented at multiple levels. CDN edge caching stores popular videos near users. Application caching using Redis or Memcached stores frequently requested metadata and user sessions. Browser caching reduces repeated downloads of static files. Together, these techniques reduce server workload, decrease buffering, and improve response time during high-demand events.

2. Multi-Cloud Container Deployment:

Solution:

The startup should package applications using Docker containers and deploy them with Kubernetes across multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud Platform. Infrastructure as Code tools such as Terraform can automate deployments and maintain consistent configurations across all environments.

Service Discovery and Scaling:

Kubernetes Services and DNS provide automatic service discovery so containers can communicate regardless of their location. Horizontal Pod Autoscaler increases or decreases the number of container instances based on CPU or memory utilization. Load balancers distribute requests evenly across available pods to ensure high availability and efficient resource utilization.

Benefits and Risks:

A multi-cloud strategy improves reliability, avoids vendor lock-in, and supports disaster recovery because workloads can shift between providers if one fails. However, organizations may face challenges such as higher operational complexity, increased monitoring requirements, data synchronization issues, and additional security management across different cloud platforms.

3. Backup and Disaster Recovery:

Backup and Recovery Design:

The e-learning platform should implement automated cloud-native backups, periodic snapshots, object versioning, and cross-region replication. Backups should be scheduled daily, while critical databases should use continuous replication. Versioning ensures accidentally deleted or modified files can be restored without affecting the latest data. Storing backup copies in multiple geographic regions protects data from regional outages.

RTO and RPO:

Recovery Time Objective (RTO) is achieved through automated failover mechanisms and pre-configured standby resources that allow services to restart quickly. Recovery Point Objective (RPO) is minimized by performing frequent backups and real-time data replication, ensuring only a small amount of recent data may be lost after a failure.

Business Continuity:

Business continuity requires continuous monitoring, regular backup verification, disaster recovery drills, documented recovery procedures, and automated alerting systems. Organizations should also implement role-based access control, encryption, and periodic testing to ensure backup files remain secure and recoverable whenever an unexpected failure occurs.

4. Hybrid Cloud Architecture:

Architecture:

Sensitive applications and confidential government data should remain in the on-premises data center, while public-facing services and non-sensitive applications can be deployed in the public cloud. A secure VPN or dedicated connection such as Direct Connect or ExpressRoute enables reliable communication between both environments while maintaining compliance requirements.

Identity Management and Encryption:

Identity and Access Management (IAM) should enforce role-based access control and multi-factor authentication to prevent unauthorized access. Data should be encrypted using strong encryption standards both during transmission (TLS/SSL) and while stored. Encryption keys should be managed using dedicated key management services for better security.

Challenges:

Managing hybrid cloud environments introduces challenges such as network latency, integration complexity, policy enforcement, security consistency, compliance monitoring, and higher operational costs. Centralized monitoring, automation tools, and regular audits help overcome these challenges and improve overall system reliability.

5. SaaS High Availability:

Multi-Region Deployment:

The SaaS application should be deployed across multiple cloud regions using an active-active or active-passive architecture. If one region becomes unavailable, user traffic is automatically redirected to another healthy region without service interruption.

Replication and Failover:

Databases should use synchronous or asynchronous replication depending on business requirements. Health checks, load balancers, DNS failover, and automatic failover mechanisms ensure that backup systems immediately take over when failures occur, minimizing downtime.

Monitoring and Alerting:

Cloud monitoring services should continuously monitor CPU, memory, response time, application logs, and database performance. Automated alerts through email or messaging platforms notify administrators immediately about failures. Dashboards, log analysis, and predictive monitoring enable proactive issue detection and maintain high application availability.